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PI Name:	Kaytor, Michael D. Ph.D.		
Project Title:	Evaluation of BIO 300 as a Radiation Countermeasure Against Proton Radiation Exposure		
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PI Email:	mkaytor@humaneticscorp.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	952-400-0406
Organization Name:	Humanetics Corporation		
PI Address 1:	7650 Edinborough Way, Suite 620		
PI Address 2:			
PI Web Page:			
City:	Edina	**State:**	MN
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Task Description:	Project Description: Humanetics Corporation (Humanetics) is developing BIO 300 as a radiation mitigator, for use in both clinical radiotherapy and as a radiation countermeasure. Through these efforts BIO 300 has demonstrated the capability to mitigate both the acute and delayed effects of radiation exposure. BIO 300 is currently being evaluated in a Phase Ib/IIa clinical trial for its ability to protect the normal pulmonary tissue of cancer patients from the side-effects of radiotherapy (NCT02567799). Despite this advanced development in human radioprotection, to date BIO 300 has only been validated as effective in countering the effects of photonic radiation (X-rays and gamma-rays), and thus the potential utility for NASA-related applications is currently unknown. In this National Space Biomedica Research Institute (NSBRI)-funded study, the objective was to evaluate BIO 300's potential to be used as a countermeasure against proton radiation exposure. The study was performed in collaboration with Keith Cengel, MD, PhD, a radiation oncologist at the University of Pennsylvania, and all experiments were performed in his laboratory. Together, we undertook the following specific aims: Specific Aim 1. To determine whether BIO 300 affects cellular radiosensitivity to protons with increasing LET (linear energy transfer). Clonogenic cell survival studies were used to evaluate BIO 300's radioprotective efficacy. This approach was performed in both cancerous and normal human cell lines. SA 1 Results. Three cell lines were used in the study, H1299 and H460 (lung cancer cell lines), and MRC5 (normal fibroblast). None of these cells exhibited significant radioprotection by BIO 300 against proton or X-ray radiation. BIO 300 did radiosensitize the H1299 cancer cells to high LET proton radiation, but not X-rays, which implicates a specific bio-interaction between BIO 300 and proton radiation, and a potential use for a BIO 300 + proton radiotherapy of specific kinds of cancer cells. Specific Aim 2. To determine whether BIO 300 affects radiation toxicities in animals exposed to protons with increasing LET. C57B6 mice, pretreated with or without BIO 300, were irradiated with protons having low or high LET to create similar whole body dose distributions. For total body doses ranging from approximately 0.5-6 Gy, we evaluated the effects of BIO 300 on whole blood cell counts, as well as DNA damage in organs. Organs of interest were lung and cardiac tissues. A cytokine panel for inflammatory markers was used to assess BIO 300's effect on mitigating radiation-induced inflammation. SA 2 Results. For technical reasons, we opted to evaluate one sub-cutaneous injection of BIO 300 in lieu of administering several oral doses as was originally intended. The whole blood counts did not yield any statistically significant differences between BIO 300 and vehicle treated animals exposed to proton radiation. Many endpoints did trend toward a protective effect by BIO 300 treatment, especially in the 0.5 Gy treated animals. Western blots of tissues extracted from animals exposed to 6 Gy high LET protons showed in many animals there were lower levels of p53 and gH2AX, a marker for DNA damage, especially in cardiac tissue. Cytokine panels showed a moderation of some specific cytokine responses over a 12 day period. Some cytokines were unaffected by BIO 300 pretreatment. Impact of results: The results of this initial proof-of-concept study warrant future investigation in a pivotal efficacy study. The lack of any radioprotective efficacy observed in the clonogenic studies could be attributed to any caveats of cell culture studies. The radiosensitization of H1299 cells implied that some NSCLC (non-small cell lung cancer) cancers may be radiosensitized by BIO 300. In the animal experiments, we observed many trends toward radioprotection by BIO 300; however, most differences were not statistically significant. We believe these trends should be further studied, as a suboptimal dosing regimen to prevent or mitigate the effects of proton radiation exposure may have been used in the current studies. Proposed future research: We propose to further investigate the route of BIO 300 administration in animal studies, as oral administration is our preferred route for future development. Chronic oral dosing may have distinct consequences from acute sub-cutaneous administration, and may explain some endpoints appearing significantly more protective than others. For example, in some western blots of tissues from BIO 300 treated animals (before exposure to 6 Gy protons) there is a striking radioprotective phenotype, but this was not correlated to a statistically significant difference in whole blood counts. To address this, we propose 60 day survival studies in mice treated with BIO 300. Overall survival will be the primary endpoint in these studies. We would additionally propose experiments of DNA damage and mutation at more mission-relevant doses of BIO 300.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Humanetics' primary commercial interest is to develop BIO 300 as an adjunct therapy to radiation therapy or chemoradiotherapy of solid tumor cancers. It functions as a protectant of normal tissue, but does not protect tumor cells from radiation therapy, and has been demonstrated to radiosensitize solid tumors. There are many potential cancer populations that would benefit from a radioprotectant like BIO 300 including, but not limited to, non-small cell lung, prostate, and head and neck cancers. BIO 300 could be easily integrated into the clinic, as it would not alter the current standard of care or be disruptive. Radiation oncologists could prescribe BIO 300 to patients during treatment planning. BIO 300 dosing would continue throughout radiotherapy with daily oral dosing performed at home by the patient. BIO 300 is being developed to address the urgent need for an effective treatment to help protect against mucositis, pneumonitis and fibrosis, mouth and throat changes, erectile dysfunction, and other side effects of radiotherapy. In addition to protection of normal tissue, BIO 300 has the potential to be used as a radiosensitizer in the treatment of a variety of solid tumor cancers enhancing radiotherapy's therapeutic effect. Toward commercializing BIO 300 in oncology, Humanetics has initiated a Phase I/IIa clinical trial of BIO 300 to determine its protective effect on normal lung tissue in non-small cell cancer patients receiving chemoradiotherapy (NCT02567799). We aim to complete this trial in 2017. Humanetics also has commercialization activities directed toward developing BIO 300, a radioprotective compound, to mitigate the risk of DNA damage resulting from medical diagnostic imaging procedures. The development of a low-dose radioprotectant is intended to alleviate the public health burden caused by the rising incidence of these tests, and provide assurances that patients, who could potentially benefit from such scans, will not be harmed by them. The risks associated with medical imaging procedures are real. A typical CT scan generates a skin dose in the range of 5 to 50 mGy, and absorbed doses from multiple CT scans can reach levels that, through epidemiological studies, are already known to induce a low rate of increased cancer incidence. These rates of radiation exposure are also within the order of magnitude of radiation exposure rates during time spent on the International Space Station (ISS) (~50-100 mGy / 6-month mission) as well as the doses of radiation during small-intensity solar particle events (SPEs). Because of this, we are interested in studies that investigate the efficacy of our product in mitigating the damage from low doses of radiation exposure. This

	is one reason for studying such a large range (both very low and very high) of doses of proton radiation exposure in our current study. BIO 300 is intended to reduce the amount of radiation-injury resulting from diagnostic medical imaging. Humanetics has collected non-clinical data demonstrating radioprotective efficacy when administered either prophylactically or post-radiation exposure. This implies that to achieve maximum radioprotection, BIO 300 should be taken before, during, and after radiation exposure associated with a diagnostic medical imaging procedure. In addition to these commercial development activities, Humanetics is developing BIO 300 as a medical countermeasure (MCM) against acute radiation syndrome (ARS) and the delayed effects of acute radiation exposure (DEARE). The specific area of DEARE interest is pneumonitis and fibrosis of the lung (DEARE-lung). BIO 300 as a medical countermeasure has been extensively investigated at the Armed Forces Radiobiology Research Institute in rodent models.
Task Progress:	Specific Aim 1: To determine whether BIO 300 affects cellular radiosensitivity for protons with increasing LET. Experimental approach: H1299, H460, and MRC5 cells were treated with 5uM BIO 300 24 hours before exposure to varying doses of proton (with different LETs) or X-rays. 1. Clonogenic cell survival studies – performed and reported in main findings (Task Description above) 2. Perform gH2AX immunostaining – not performed, animal experiments were prioritized 3. Perform western blots of DNA damage – not performed, animal experiments were prioritized Specific Aim 2: To determine whether BIO 300 affects radiation toxicities in animals exposed to protons with increasing LET. Experimental approach: C57B6 mice, pretreated with or without BIO 300, were irradiated with protons having low or high LET to create similar whole body dose distributions 1. Whole blood cell counts – performed and reported in main findings (Task Description above) 2. Western blots of DNA damage in tissues – performed and reported in main findings (Task Description above) 3. Luminex panels for markers of inflammation – performed and reported
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